

Calculus III

"Exercises"

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Multiple Integration

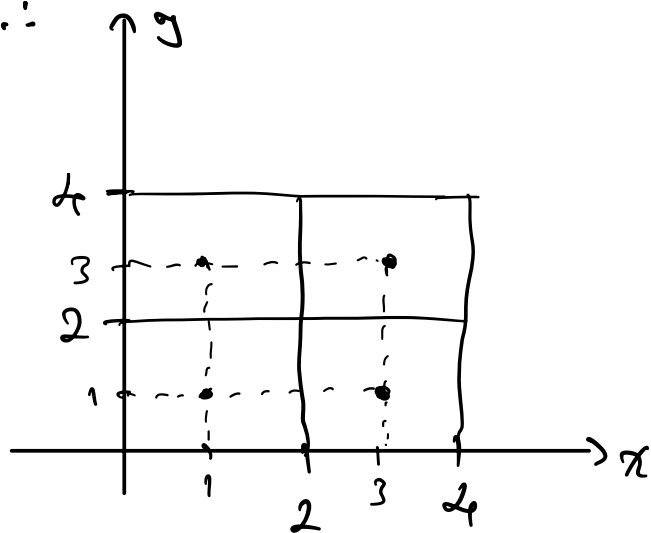
- 14.1 Double Integration -

② Let D be a region in the xy -plane given by the inequalities $0 \leq x \leq 4$, $0 \leq y \leq 4$. Consider the partition P obtained by dividing the square D into four equal subsquares. Approximate the double integral

$$\iint_D (x^2 + y) \, dA$$

i) by computing the Riemann sum $R(f, P)$ using the center point of each subsquare

Sol.:



Let P be the partition of D where each rectangle element σ_{ij} is a square with length 2. Then for each $i, j = \{1, 2\}$ the area element ΔA_{ij} has area 4. Choose the midpoints (x_{ij}^*, y_{ij}^*) of each σ_{ij} . Then,

$$\begin{aligned} \iint_D (x^2 + y) \, dA &\approx R_c(f, P) = \sum_{i=1}^2 \sum_{j=1}^2 f(x_{ij}^*, y_{ij}^*) \Delta A_{ij} \\ &= [f(1, 1) + f(1, 3) + f(3, 1) + f(3, 3)] \cdot 4 \\ &= 120. \end{aligned}$$

□

ii) by computing the lower Riemann sum corresponding to the partition P .

Sol.: $R_L(f, P) = \sum \sum m_f \{f(x_{ij}^*, y_{ij}^*)\} \Delta A_{ij}$. Since $f(x, y)$ and $f(0, y)$ are continuous and monotone for $x, y \geq 0$, choosing $m_f \{f\}$ for each rectangle implies that choosing $m_f \{f\}$ on that rectangle. Plus,

$$R_L(f, P) = [f(0,0) + f(0,2) + f(2,0) + f(2,2)] \cdot 4 = 48 \quad \square$$

iii) by computing the upper Riemann sum corresponding to the partition P .

$$\begin{aligned} R_u(f, P) &= \sum \sum \sup \{f(x_{ij}^*, y_{ij}^*)\} \Delta A_{ij} \\ &= [f(2,2) + f(2,4) + f(4,2) + f(4,4)] \cdot 4 = 208 \quad \square \end{aligned}$$

② Evaluate the following integrals by inspection.

i) $\iint_D (x+3) dA$, where D is the upper semi circle with radius 2.

Sol.:
$$\iint_D (x+3) dA = \underbrace{\iint_D x dA}_{=0} + 3 \iint_D dA = 3 \cdot \pi \frac{2^2}{2} = 6\pi. \quad \square$$

 $= 0$, since D is symmetric about y -axis and $f(x) = x$ is odd.

ii) $\iint_D \sqrt{a^2 - x^2 - y^2} dA$, where $D = \{(x, y) : x^2 + y^2 \leq a^2\}$.

Sol.: we are asked to evaluate the area under the hemisphere with radius a over a circle with the same radius. This is actually the volume of a hemisphere with radius a . Thus, $\frac{1}{2} \cdot \left(\frac{4}{3} \pi a^3\right) = \frac{2}{3} \pi a^3. \quad \square$

$$\text{iii) } \iint_D a - \sqrt{x^2 + y^2} \, dA, \text{ where } D = \{(x, y) : x^2 + y^2 \leq a^2\}.$$

Sol.: Note that this is the volume of a right circular cone with height a . $\square$
 So, $\frac{1}{3}(\pi a^2)(a) = \frac{\pi}{3}a^3$.

$$\text{iv) } \iint_D x + y \, dA, \text{ where } D \text{ is a parallelogram with vertices } (2, 2), (1, -1), (-2, -2), (-1, 1).$$

Sol.: Note that vertices of D is symmetric with respect to the origin, so D is centrally-symmetric. Also note that $f(x, y) = x + y = -((-x) + (-y)) = -f(-x, -y)$, $\square$

hence f is odd and therefore integral is equal to 0.

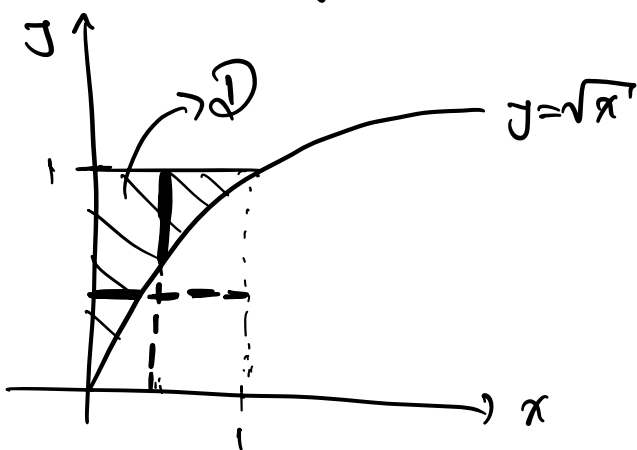
- 1dp.2 Iterated Double Integrals -

② Evaluate the given integrals

$$i) \int_0^1 \int_{\sqrt{x}}^1 e^{y^3} dy dx.$$

Sol.:

There is no solution in terms of elementary functions for the inner integral. So, we shall try to re-iterate the integral.

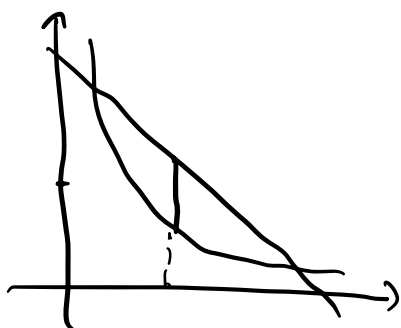


Note that the variable y is bounded below by $\sqrt{x}$ and above by 1, and $x \in [0, 1]$. So the domain D of the integration is as sketched. Now consider a horizontal line in D and describe the boundary of x in terms of y . From sketch, clearly, x is bounded below by 0 and above by y^2 , and $y \in (0, 1]$. Thus,

$$I = \int_0^1 \int_{\sqrt{x}}^1 e^{y^3} dy dx = \int_0^1 \int_0^{y^2} e^{y^3} dx dy = \int_0^1 y^2 e^{y^3} dy = \left. \frac{1}{3} e^{y^3} \right|_0^1 = \frac{e-1}{3}. \quad \square$$

ii) $\iint_D \ln x \, dA$, where D is the finite region in the first quadrant bounded by $2x+2y=5$ and $xy=2$.

Sol.:



Boundary curves meet at the points $(\frac{1}{2}, 2)$ and $(2, \frac{1}{2})$

since, $y = \frac{5}{2} - x = \frac{1}{x} \Rightarrow x^2 - \frac{5}{2}x + 1 = 0 \Rightarrow x = \frac{1}{2}, 2$.

So,

$$\begin{aligned} \iint_D \ln x \, dA &= \int_{1/2}^2 dx \int_{1/x}^{\frac{5}{2}-x} \ln x \, dy = \int_{1/2}^2 \left(\frac{5}{2} - x - \frac{1}{x} \right) \ln x \, dx = \int_{1/2}^2 \frac{5}{2} \ln x - x \ln x - \frac{\ln x}{x} \, dx \\ &= \left[\frac{5}{2} x (\ln x - 1) \right] - \left[\frac{x^2}{2} \left(\ln x - \frac{1}{2} \right) \right] - \left[\frac{\ln^2 x}{2} \right] \bigg|_{1/2}^2 = \frac{33}{8} \ln 2 - \frac{45}{16}. \quad \square \end{aligned}$$

② Find the volume of the given solids.

i) Under $z = 1 - x^2 - y^2$ and above the region $x \geq 0, y \geq 0, x + y \leq 1$.

Sol.: From the equalities we get $0 \leq y \leq 1 - x$ and $0 \leq x \leq 1$. Thus, we shall solve

$$\int_0^1 dx \int_0^{1-x} (1 - x^2 - y^2) \, dy = \frac{1}{3}. \quad \square$$

ii) Above the xy -plane and under the surface $z = 1 - x^2 - 2y^2$.

Sol.: we shall solve the integral $\iint_{x^2 + 2y^2 \leq 1} (1 - x^2 - 2y^2) \, dA$, where the boundary comes from $z = 0$.

To translate the integral into iterated one observe that $x^2 + 2y^2 \leq 1$ implies that $|y| \leq \sqrt{\frac{1-x^2}{2}}$ and that $|x| \leq 1$. Thus,

$$\int_{-1}^1 dx \int_{-\sqrt{(1-x^2)/2}}^{\sqrt{(1-x^2)/2}} (1 - x^2 - 2y^2) \, dy = \frac{\pi}{2\sqrt{2}}. \quad \square$$

- Ex. 3 Improper Integrals -

(2) Determine whether the given integral converges or diverges.

i) $\iint_D e^{-x-y} dA$, where D is the first quadrant of the xy -plane.

Sol.:
$$\lim_{m \rightarrow \infty} \int_0^m \int_0^m e^{-x-y} dy dx \stackrel{*}{=} \lim_{m \rightarrow \infty} \int_0^m dx \lim_{m \rightarrow \infty} \int_0^m e^{-x} \cdot e^{-y} dy = 1.$$

Note that since the integrand is non-negative (ok with non-positive) we can write $*$, otherwise the safest version is that the taking limit at the end of the integration.

$$= -e^{-x} e^{-y} \Big|_0^m = -e^{-x} (e^{-m} - 1)$$

$$= e^{-x}$$

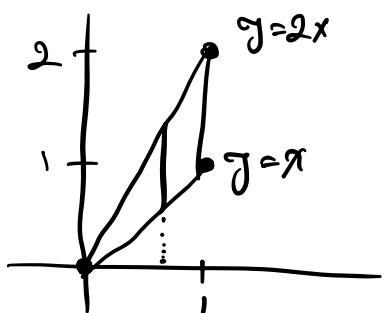
$$= -(e^{-m} - 1)$$

1

□

ii) $\iint_D \frac{1}{x\sqrt{y}} dA$ over the triangle D with vertices $(0,0), (1,1), (1,2)$.

Sol.:



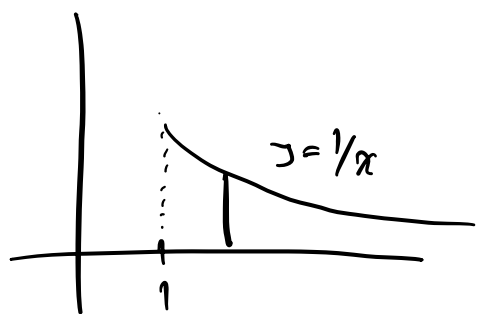
$$I = \int_0^1 dx \int_x^{2x} \frac{1}{x\sqrt{y}} dy dx,$$

$$J := \frac{1}{x} \int_x^{2x} y^{-1/2} dy = 2 \frac{1}{x} \left[\sqrt{y} \Big|_x^{2x} \right] = 2(\sqrt{2}-1) \frac{\sqrt{x}}{x}.$$

$$\Rightarrow I = 2(\sqrt{2}-1) \int_0^1 \frac{\sqrt{x}}{x} dx = 2(\sqrt{2}-1) \cdot 2\sqrt{x} \Big|_0^1 = 4(\sqrt{2}-1). \quad \square$$

iii) $\iint_{\mathcal{D}} \frac{\partial \mathcal{A}}{x+y}$, where $\mathcal{D}$ is lying above x -axis, under the curve $y = 1/x$ and to the right of the line $x=1$.

Sol.:



$$\mathcal{I} = \iint_{\mathcal{D}} \frac{\partial \mathcal{A}}{x+y} = \int_1^{\infty} 2x \int_0^{1/x} \frac{\partial y}{x+y} = \int_1^{\infty} \ln \left| 1 + \frac{1}{x^2} \right| 2x.$$

Note that for any $a > 0$, we have $0 < \ln|1+a| < a$. Thus,

$$0 < \mathcal{I} < \int_1^{\infty} \frac{1}{x^2} 2x = 2. \text{ Therefore, } \mathcal{I} \text{ converges to a number}$$

somewhere between 0 and 2.

iv) $\iint_{\mathcal{D}} \frac{1}{1+x+y} \partial y \partial x$, where $\mathcal{D} = \{(x, y) : 0 \leq x < \infty, 0 < y < 1\}$.

Sol.: Since $y < 1$, $\frac{1}{1+x+y} > \frac{1}{x+2}$. Hence, $\iint_{\mathcal{D}} \frac{1}{x+2} \partial y \partial x < \iint_{\mathcal{D}} \frac{1}{1+x+y} \partial y \partial x$. But the left hand side diverges. Therefore, our integral diverges. $\square$

② Find the average value of the given function over the given region.

i) $x^2 + y^2$ over the triangle $\mathcal{T}$ $0 \leq x \leq a$, $0 \leq y \leq a-x$.

Sol.: f is continuous on the closed, bounded, connected set $\mathcal{T}$. Thus, by Mean Value

Theorem, there exists a point (x_0, y_0) such that

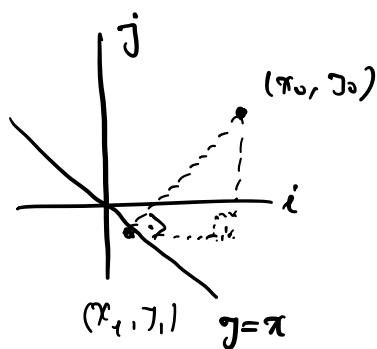
$$\iint_{\mathcal{T}} f(x, y) \partial \mathcal{A} = f(x_0, y_0) \cdot \mathcal{A},$$

where $\mathcal{A}$ is the area of the set $\mathcal{T}$. Then the average is found by $\frac{1}{\mathcal{A}} \iint_{\mathcal{T}} f \partial \mathcal{A}$. $\square$

ii) Find the average distance from points in the quarter disk $x^2 + y^2 \leq a^2$, $x \geq 0$, $y \geq 0$ to the line $x + y = 0$.

Sol.:

Let (x_0, y_0) be an arbitrary point on the xy -plane and (x_1, y_1) the closest point to the point (x_0, y_0) on the line. Then the distance d is given by



$$d^2 = (y_0 - y_1)^2 + (x_0 - x_1)^2$$

$$= 2(x_0 - x_1)^2$$

$$\left[\text{since } \frac{y_0 - y_1}{x_0 - x_1} = 1 \right]$$

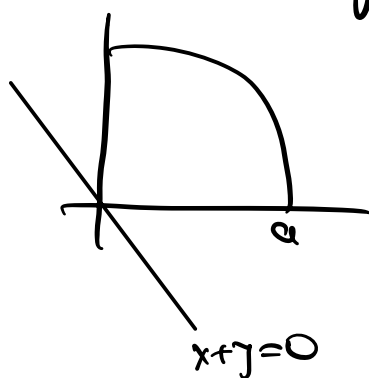
$$= 2\left(x_0 - \frac{x_0 + y_0}{2}\right)^2$$

$$\left[\text{since } \frac{y_0 + x_1}{x_0 - x_1} = 1 \Leftrightarrow x_1 = -y_1 \right]$$

$$= \frac{(x_0 + y_0)^2}{2}$$

$$\Rightarrow d = \frac{|x_0 + y_0|}{\sqrt{2}}$$

Then the average distance is



$$\bar{d} = \frac{1}{\frac{1}{4}\pi a^2} \int_0^a \int_0^{\sqrt{a^2 - x^2}} \frac{x+y}{\sqrt{2}} dy dx = \frac{4\sqrt{2}}{3\pi} a.$$

□

Find area of $x^2 + y^2 = 1$ using double integral, it's important for what I suffer when expressing volume by double integral.

— 12.4 In Polar Coordinates —

② Compute $\int_{\mathbb{R}} e^{-x^2} dx$.

Sol.:

Let $I := \int_{\mathbb{R}} e^{-x^2} dx$. Then,

$$I^2 = \left(\int_{\mathbb{R}} e^{-x^2} dx \right)^2 = \int_{\mathbb{R}} e^{-x^2} dx \cdot \int_{\mathbb{R}} e^{-y^2} dy$$

$$= \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA$$

$$= \int_0^{2\pi} d\theta \int_0^{\infty} e^{-r^2} r dr$$

$$= 2\pi \cdot \lim_{R \rightarrow \infty} \left. \frac{-1}{2} e^{-r^2} \right|_0^R = \pi.$$

□

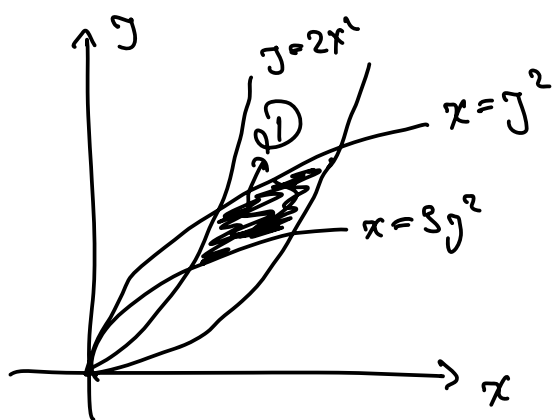
Hence, $I = \sqrt{\pi}$.

② Find the area of the finite plane region bounded by the four parabolas $y=x^2$, $y=2x^2$, $x=y^2$, $x=3y^2$.

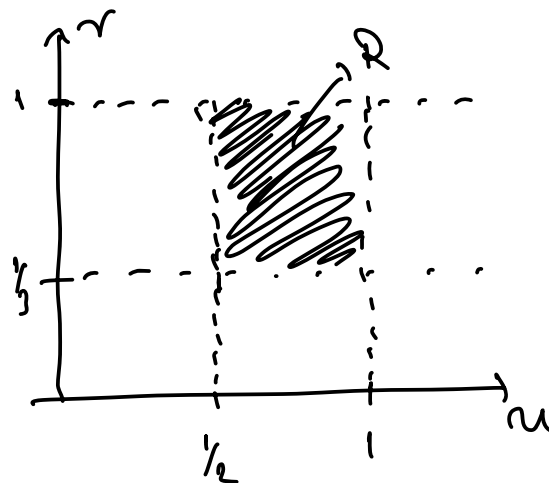
Sol.:

Let D be the region that we are interested in. Let $u = \frac{x^2}{y}$ and $v = \frac{y^2}{x}$.

Then the region D in xy -plane corresponds to the rectangle R in the uv -plane



$$\begin{aligned} u &= x^2/y \\ v &= y^2/x \end{aligned}$$



The Jacobian of the transformation is

$$\left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{\left| \frac{\partial(u, v)}{\partial(x, y)} \right|} = \frac{1}{\begin{vmatrix} 2x/y & -x^2/y^2 \\ -y^2/x^2 & 2y/x \end{vmatrix}} = \frac{1}{3} \quad \left(\text{We take the absolute value of the determinant.} \right)$$

Hence,

$$\iint_D dx dy = \iint_R \frac{1}{3} du dv = \frac{1}{3} \left(\frac{1}{2} \cdot \frac{2}{3} \right) = \frac{1}{9}.$$

□

- H. 5 Graph Integrals -

(Lecture)

- 14.6 Change of Variables in Triple Integrals -

(Later)

- 12.1 Vector Functions of One Variable -

② Describe the motion of a particle whose position at time t is given by

$$i) \vec{r}(t) = t \hat{i} + t^2 \hat{j} + t^3 \hat{k} \text{ at } (1, 1, 1).$$

Sol.:

Since the scalar parametric equations for the curve are $x=t$, $y=t^2$, and $z=t^3$ which satisfies $y=x^2$ and $z=x^3$, the curve is the curve of intersection of the two cylinders $y=x^2$ and $z=x^3$. And

$$\vec{v}_{(1,1,1)} = \hat{i} + 2t\hat{j} + 3t^2\hat{k} \Big|_{(1,1,1)} = \hat{i} + 2\hat{j} + 3\hat{k},$$

$$\vec{a}_{(1,1,1)} = 2\hat{j} + 6t\hat{k} \Big|_{(1,1,1)} = 2\hat{j} + 6\hat{k}.$$

Note that $\vec{v}$ is always tangent to the path of motion at point $\vec{r}(t)$. $\square$

$$ii) \vec{r}(t) = 3\cos(\omega t) \hat{i} + 4\cos(\omega t) \hat{j} + 5\sin(\omega t) \hat{k}.$$

Sol.:

Note that $x^2 + y^2 + z^2 = 25$, hence the path of the particle lies on this sphere.

Also since $x=3\cos(\omega t)$ and $y=4\cos(\omega t)$, the path of the particle also lies on the plane $4x=3y$. Therefore, the exact path is the intersection of these two surfaces. $\square$

— 11.3 Curves and Parametrizations —

① Parametrize the curve of intersection of a given surfaces

i) $S_1: x+2y+4z=4$, $S_2: x^2+4y^2=4$.

Sol.:

Obviously, parametrization is not unique, in general. Also note that S_2 is independent of z . Thus, it seems one convenient way is to write $x=2\cos t$, $y=\sin t$ for $0 \leq t \leq 2\pi$. Solving for z then gives us $z=1-\frac{1}{2}(\cos t + \sin t)$. Hence,

$$\vec{r}(t) = 2\cos t \hat{i} + \sin t \hat{j} + \left(1 - \frac{\cos t + \sin t}{2}\right) \hat{k}, \quad 0 \leq t \leq 2\pi$$

is one of the parametric representation of the curve of intersection of S_1 and S_2 . □

ii) $S_1: x^2+y+z=2$, $S_2: xy+z=1$.

Sol.:

Neither of them are independent of a variable. But we can obtain a third surface S_3 that contains the curve of intersection of S_1 and S_2 by subtracting $S_1 - S_2$:

$$S_3: x^2+y-xy=1.$$

Then let $x=t$, hence

$$\vec{r}(t) = t\hat{i} + (1+t)\hat{j} + (1-t-t^2)\hat{k}.$$

□

② Consider the parametric curve $\vec{r}(t) = a\cos t \hat{i} + a\sin t \hat{j} + bt \hat{k}$.

i) Find the length s of that curve between the points $(a,0,0)$ and $(a,0,2\pi b)$.

Sol.: The required length s corresponds to parameter interval $[0, 2\pi]$. So,

$$s = \int_0^{2\pi} \left\| \frac{d\vec{r}}{dt} \right\| dt = \int_0^{2\pi} \left[(-a\sin t \hat{i} + a\cos t \hat{j} + b \hat{k}) \cdot (-a\sin t \hat{i} + a\cos t \hat{j} + b \hat{k}) \right]^{1/2} dt = \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi \sqrt{a^2 + b^2}.$$

□

ii) Reparametrize $\vec{r}(t)$ in terms of the arc-length measured from point $(a, 0, 0)$ in the direction of increasing t .

Sol.:

Since $s(t) = \int_0^t \sqrt{a^2 + b^2} \, d\tau = t\sqrt{a^2 + b^2}$, $t(s) = s/\sqrt{a^2 + b^2}$. Hence,

$$\vec{r}(s) = \vec{r}(t(s)) = a \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \hat{i} + a \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \hat{j} + b \frac{s}{\sqrt{a^2 + b^2}} \hat{k}.$$

□

- 15.1 Vector and Scalar Fields -

② Find the field lines of

$$i) \vec{F} = xz \hat{i} + 2x^2z \hat{j} + x^2 \hat{k}.$$

Sol.:

The field lines satisfy

$$\frac{dx}{xz} = \frac{dy}{2x^2z} = \frac{dz}{x^2}.$$

We then only need to solve the system of differential equations:

$$dy = 2x dx \Rightarrow y = x^2 + C_1$$

$$dy = 2z dz \Rightarrow y = z^2 + C_2.$$

Therefore, the field lines are the curves of intersection of two "families",
 $y = x^2 + C_1$ and $y = z^2 + C_2$. □

$$ii) \vec{F}(r, \theta) = \hat{r} + 2\hat{\theta}.$$

Sol.: In polar coordinates $\vec{F}$ satisfies

$$\frac{dr}{1} = \frac{r d\theta}{2}.$$

□

— 15.2 Conservative Fields —

- ① Show that the gravitational field $\vec{F}(\vec{r}) = -km \frac{\vec{r} - \vec{r}_0}{\|\vec{r} - \vec{r}_0\|^3}$ is conservative by showing that $\phi(x, y, z) = \frac{km}{\|\vec{r} - \vec{r}_0\|}$ is a potential for $\vec{F}$.

Sol.:

Just check that

$$\frac{\partial \phi}{\partial x} = F_1(\vec{r}), \quad \frac{\partial \phi}{\partial y} = F_2(\vec{r}), \quad \frac{\partial \phi}{\partial z} = F_3(\vec{r}).$$

□

- ② Show that the vector field $\vec{F} = (xy - \sin z)\hat{i} + (\frac{1}{2}x^2 - e^y/z)\hat{j} + (e^y/z^2 - x \cos z)\hat{k}$ is conservative in $D = \{(x, y, z) : z \neq 0\}$. Find a potential for $\vec{F}$.

Sol.:

First, note that

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} = x, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} = -\cos z, \quad \text{and} \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y} = e^y/z^2.$$

Hence, $\vec{F}$ might be conservative. To be sure that we need to find a potential ϕ for $\vec{F}$.

Suppose ϕ is a potential for $\vec{F}$. Then,

$$\frac{\partial \phi}{\partial x} = xy - \sin z \Rightarrow \phi = \frac{1}{2}x^2y - x \sin z + C_1(y, z),$$

$$\frac{\partial \phi}{\partial y} = \frac{1}{2}x^2 - \frac{e^y}{z} \Rightarrow \phi = \frac{1}{2}x^2y - \frac{e^y}{z} + C_2(x, z),$$

$$\frac{\partial \phi}{\partial z} = \frac{e^y}{z^2} - x \cos z \Rightarrow \phi = -\frac{e^y}{z} - x \sin z + C_3(x, y).$$

Thus, $\phi(x, y, z) = \frac{1}{2}x^2y - x \sin z - \frac{e^y}{z} + C$. And hence $\vec{F}$ is conservative on D . □

- 15.3 Line Integrals -

① Compute $\int_C x^2 + y^2 ds$, where C is the straight line from the origin to the point $(2, 1)$.

Sol.:

C can be parametrized by $x=2t$, $y=t$ for $0 \leq t \leq 1$. That is,

$$C: \vec{r}(t) = 2t\hat{i} + t\hat{j}, \quad 0 \leq t \leq 1.$$

Hence, the arc-element ds becomes $ds = \left\| \frac{d\vec{r}}{dt} \right\| dt = \|2\hat{i} + \hat{j}\| dt = \sqrt{5} dt$.

Thus,

$$\int_C x^2 + y^2 ds = \int_0^1 (4t^2 + t^2) \sqrt{5} dt = 5\sqrt{5}/3. \quad \square$$

② Find the centroid of the circular helix C given by $\vec{r} = a \cos t \hat{i} + a \sin t \hat{j} + bt \hat{k}$, $0 \leq t \leq 2\pi$.

Sol.:

Let L be the length of the C and M_{x_i} be the moment about the axis $x_i = 0$. To find the x_i component of the centroid we need to compute M_{x_i}/L . So,

$$L = \int_C ds = \int_0^{2\pi} \|-a \sin t \hat{i} + a \cos t \hat{j} + b \hat{k}\| dt = \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi \sqrt{a^2 + b^2}.$$

$$M_z = \int_C z ds = \int_0^{2\pi} bt \sqrt{a^2 + b^2} dt = 2\pi^2 b \sqrt{a^2 + b^2}.$$

$$M_x = M_y = 0.$$

Therefore, the centroid of C is the point $(0, 0, \pi b)$. $\square$

- 15.4 Line Integrals of Vector Fields -

① Let $\vec{F} = y^2 \hat{i} + 2xy \hat{j}$. Compute $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(2,2)$ along

i) $y=x$.

Sol.:
In that case, $\vec{r} = t\hat{i} + t\hat{j}$, $0 \leq t \leq 2$. Hence $d\vec{r} = (\hat{i} + \hat{j})dt$. Thus,

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 (t^2 \hat{i} + 2t^2 \hat{j}) \cdot (\hat{i} + \hat{j}) dt = \int_0^2 3t^2 dt = 2. \quad \square$$

ii) $y=x^2$.

Sol.:
In that case, $\vec{r} = t\hat{i} + t^2\hat{j}$, $0 \leq t \leq 2$. Hence $d\vec{r} = (\hat{i} + 2t\hat{j})dt$. Thus,

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 (t^4 \hat{i} + 2t^3 \hat{j}) \cdot (\hat{i} + 2t\hat{j}) dt = \int_0^2 5t^4 dt = 2. \quad \square$$

iii) the piecewise smooth path consisting of the straight line segments from $(0,0)$ to $(0,2)$ and from $(0,2)$ to $(2,2)$.

Sol.:

In that case, since C is not a smooth curve but consists of two smooth curves, we need to compute

$$\int_C \vec{F} \cdot d\vec{r} = \sum_{i=1}^2 \int_{C_i} \vec{F} \cdot d\vec{r},$$

where C_1 and C_2 are the straight line segments from $(0,0)$ to $(0,2)$ and from $(0,2)$ to $(2,2)$, respectively. We shall feel free to parametrize each separately. Thus,

$$\int_C \vec{F} \cdot d\vec{r} = \sum_{i=1}^2 \int_{C_i} \vec{F} \cdot d\vec{r},$$

$$\begin{aligned}
&= \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} \\
&= \int_0^1 (t^2 \hat{i}) \cdot (\hat{j}) dt + \int_0^1 (\hat{i} + 2t\hat{j}) \cdot (\hat{i}) dt \\
&= 0 + 1 \\
&= 1.
\end{aligned}$$

□

(2) Consider the vector field

$$\vec{F} = A\pi \sin(\pi y) \hat{i} + (x^2 \cos(\pi y) + Bye^{-z}) \hat{j} + y^2 e^{-z} \hat{k},$$

where A and B are constants.

i) For what values of the constants A and B is the vector field F conservative?

Sol.:

By necessary condition, F cannot be conservative unless

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \quad \text{and} \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}.$$

So, we require that

$$A\pi x \cos(\pi y) = 2x \cos(\pi y), \quad 0 = 0, \quad \text{and} \quad -Bye^{-z} = 2ye^{-z}.$$

Hence, $A = 2/\pi$ and $B = -2$. In that case one can find that $F = \nabla \phi$, where

$$\phi = \frac{x^2 \sin(\pi y)}{\pi} - y^2 e^{-z}.$$

□

ii) Compute $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve $\vec{r} = \cos t \hat{i} + \sin 2t \hat{j} + \sin^2 t \hat{k}$ for $0 \leq t \leq 2\pi$.

Sol.:

Since F is conservative and ϕ is a potential for F , we have

$$I = \int_C \vec{F} \cdot d\vec{r} = \int \partial\phi = \phi(t_1) - \phi(t_0),$$

where t_0, t_1 are the initial and terminal points of C . Also note that $t_0 = t_1$, since $r(0) = r(2\pi) = \hat{i}$. Thus, $I = 0$. $\square$

iii) Compute $\int_C \vec{F} \cdot d\vec{r}$, where C is the curve of intersection of the paraboloid $z = x^2 + 4y^2$ and the plane $z = 3x - 2y$ from the origin to the $(1, 1/2, 2)$.

Sol.:

Once again we have

$$\int_C \vec{F} \cdot d\vec{r} = \phi(t_1) - \phi(t_0).$$

But, this means the result of the left-hand side is independent of choice of C . Thus, without any consideration of C ,

$$\int_C \vec{F} \cdot d\vec{r} = \frac{x^2 \sin(\pi y)}{\pi} - y^2 e^{-x} \Big|_{(0,0,0)}^{(1, 1/2, 2)} = \frac{1}{\pi} - \frac{1}{4e^2}. \quad \square$$

③ Compute $I = \int_C (e^x \sin y + 3y) dx + (e^x \cos y + 2x - 2y) dy$ counter-clockwise around the ellipse $4x^2 + y^2 = 4$.

Sol.:

We can re-write the integral as

$$I = \int_C \vec{F} \cdot d\vec{r},$$

where $\vec{F} = (e^x \sin y + 3y) \hat{i} + (e^x \cos y + 2x - 2y) \hat{j}$. Now, simply note that if the $3y$ term were $2y$, then $\vec{F}$ would be a conservative field. Hence, $\vec{F}$ is indeed of the form

$$\vec{F} = \nabla\phi + y \hat{i},$$

which consists of a conservative part and non-conservative part. Then,

$$I = \oint_C \nabla \phi \cdot d\vec{r} + \oint_C g dx.$$

The first term of the right hand-side is 0 since $\nabla \phi$ is conservative and C is closed. For the second term, let $x = \cos t$, $y = 2 \sin t$ where $0 \leq t \leq 2\pi$ be the parametrization of C . Then,

$$I = 0 + \oint_C g dx = \int_0^{2\pi} (2 \sin t) \cdot (-\sin t) dt = -2\pi.$$

□

- 15.5 Surface Integrals -

① Compute $\iint_S z \, dS$ over the conical surface $z = \sqrt{x^2 + y^2}$ between $z=0$ and $z=1$.

Sol.:
The surface S can be represented by $\vec{r} = x\hat{i} + y\hat{j} + \sqrt{x^2 + y^2}\hat{k}$. Then the surface element becomes

$$\begin{aligned} dS &= \left\| \frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} \right\| dx dy = \left\| \left(\hat{i} + \frac{x}{z}\hat{k}\right) \times \left(\hat{j} + \frac{y}{z}\hat{k}\right) \right\| dx dy = \left\| -\frac{x}{z}\hat{i} - \frac{y}{z}\hat{j} + \hat{k} \right\| dx dy \\ &= \sqrt{1 + \frac{x^2}{z^2} + \frac{y^2}{z^2}} dx dy = \sqrt{2} dx dy. \end{aligned}$$

Hence,

$$\iint_S z \, dS = \iint_{x^2+y^2 \leq 1} z \sqrt{2} dx dy = \int_0^{2\pi} \int_0^1 r^2 dr d\theta = \frac{2\sqrt{2}}{3} \pi.$$

□

② Use polar coordinates in the uv -plane to find the moment of inertia of the parametric surface S given by $x=2uv$, $y=u^2-v^2$, $z^2=u^2+v^2$, where $u^2+v^2 \leq 1$.

Sol.:

Set $u := r \cos \theta$, $v := r \sin \theta$. Hence,

$$x = r^2 \sin(2\theta), \quad y = r^2 \cos(2\theta), \quad z = r^2 \quad \text{with } 0 \leq \theta \leq \pi \text{ and } r \leq 1.$$

Then the surface element becomes

$$\begin{aligned} dS &= \left\| \frac{\partial \vec{r}}{\partial r} \times \frac{\partial \vec{r}}{\partial \theta} \right\| dr d\theta = \sqrt{\left(\frac{\partial(z,y)}{\partial(r,\theta)}\right)^2 + \left(\frac{\partial(y,z)}{\partial(r,\theta)}\right)^2 + \left(\frac{\partial(z,x)}{\partial(r,\theta)}\right)^2} dr d\theta \\ &= \sqrt{(-4r^3)^2 + (4r^3 \sin(2\theta))^2 + (4r^3 \cos(2\theta))^2} dr d\theta \\ &= 4\sqrt{2} r^3 dr d\theta. \end{aligned}$$

Hence, the moment of inertia about the z -axis is

there might be a mistake

$$\iint_S z^2 dS = 4\sqrt{2} \int_0^{\pi} \int_0^1 r^7 dr d\theta = \pi/\sqrt{2}.$$

□

(3) Compute the moment about $z=0$, that is $\iint_S z dS$, where S is the hyperboloid $z^2 = 1 + x^2 + y^2$ between the planes $z=1, z=\sqrt{5}$.

Sol:

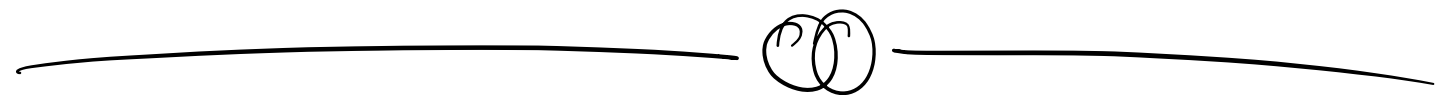
In this problem S is given by the equation $\phi(x, y, z) = 1 + x^2 + y^2 - z^2 = 0$. Note that $\nabla\phi$ is a non-zero normal vector to the surface element at (x, y, z) and that $\hat{k} \cdot \vec{n} = \partial_z \phi$. Hence, without parametrization, the surface element can be written as

$$\begin{aligned} dS &= \left\| \frac{\nabla\phi}{\partial_z \phi} \right\| dx dy \\ &= \left\| \frac{2x\hat{i} + 2y\hat{j} - 2z\hat{k}}{-2z} \right\| dx dy \\ &= \frac{\sqrt{x^2 + y^2 + z^2}}{z} dx dy = \frac{\sqrt{1 + 2(x^2 + y^2)}}{z} dx dy. \end{aligned}$$

Thus,

$$\iint_S z dS = \iint_{x^2 + y^2 \leq 4} \sqrt{1 + 2(x^2 + y^2)} dx dy = \int_0^{2\pi} \int_0^2 \sqrt{1 + 2r^2} r dr d\theta = \frac{26}{3} \pi.$$

□



* Parameterization of Curves - w/6*

(1a) Parameterization is not a unique process. But here is a standard way of doing that

Let C be a circle centered at (h, k) with radius r . If the orientation is counter clockwise we choose " t " hence " $+ \sin t$ ". Otherwise we choose " $-t$ " hence " $-\sin t$ ".

Plus, set

$$x(t) = h + r \cos t = 2 + 2 \cos t \quad y(t) = k - r \sin t = -2 \sin t.$$

Also, the interval of t

$$(4, 0): \quad x(t_{\min}) = 2 + 2 \cos(t_{\min}) = 4, \quad \Rightarrow t_{\min} = 0 \\ y(t_{\min}) = -2 \sin(t_{\min}) = 0$$

$$(2, 2): \quad x(t_{\max}) = 2 + 2 \cos(t_{\max}) = 2, \quad \Rightarrow t_{\max} = \frac{3}{2}\pi. \\ y(t_{\max}) = -2 \sin(t_{\max}) = 2$$

Therefore,

$$C: (2 + 2 \cos t, -2 \sin t), \text{ where } 0 \leq t \leq \frac{3}{2}\pi.$$

□

(1b) Set $x = \cos t$, $y = \sin t$. Then, $z = 2 \sin t + 1$. Plus, $0 \leq t \leq 2\pi$. Hence,

$$C: \vec{r}(x, y, z) = \cos t \hat{i} + \sin t \hat{j} + (2 \sin t + 1) \hat{k}, \text{ for } 0 \leq t \leq 2\pi.$$

(1c) Set $x = t$. Then $y = 1 - t$ and $z = \sqrt{2t(1-t)}$. And since $z \geq 0$, $0 \leq t \leq 1$.

(1d) Note that we have $4 - x^2 - z^2 = 4 - 2x^2$. Hence, $z = \pm x$. So there will be two ℓ_1, ℓ_2 for $z = x$ and $z = -x$, respectively. And C is the "union" of ℓ_1 and ℓ_2 . Hence, set $x = t$, then

$$\ell_1(t) = t \hat{i} + (4 - 2t^2) \hat{j} + t \hat{k}$$

$$\ell_2(t) = t \hat{i} + (4 - 2t^2) \hat{j} - t \hat{k}.$$

There is no restriction on t . Hence $t \in (-\infty, \infty)$.

□

$$\textcircled{2} \quad \mathcal{L} = \int_{t_1}^{t_2} \left\| \frac{d\vec{r}}{dt} \right\| dt = \int_{-\frac{\pi}{2}}^{\pi} \|t \cos t \hat{i} + t \sin t \hat{j}\| dt = \int_{-\frac{\pi}{2}}^{\pi} |t| dt = \int_{-\frac{\pi}{2}}^0 -t dt + \int_0^{\pi} t dt = \frac{5}{2} \pi^2. \quad \square$$

3a) We interpret the arc-length element ds as dl . This is a scalar line integral over a parametric curve, with integrand $f(x, y, z) = x - y + 2z$. So,

$$\int_C f(x, y, z) dl = \int_C f(\vec{r}(t)) \left\| \frac{d\vec{r}}{dt} \right\| dt, \text{ where}$$

$$f(\vec{r}(t)) = (1+t) - (2t) + 2(1-t) = 3(1-t), \text{ and}$$

$$dl = \left\| \frac{d\vec{r}}{dt} \right\| dt = \|\hat{i} + 2\hat{j} - \hat{k}\| dt = \sqrt{6} dt.$$

Thus, the given integral is equivalent to the

$$3\sqrt{6} \int_0^1 (1-t) dt = \frac{3}{2} \sqrt{6}. \quad \square$$

3b) Similar to the previous one:

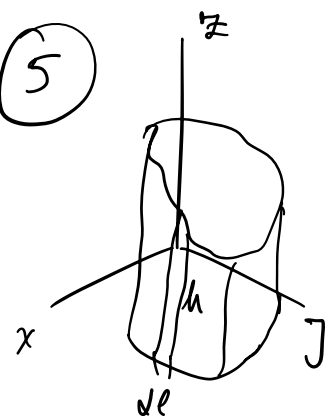
$$\begin{aligned} \int_C f(\vec{r}(t)) dl &= \int_C (\cos^2 t + \sin^2 t + t) \|- \sin t \hat{i} + \cos t \hat{j} + \hat{k}\| dt = \int_0^1 (1+t) \sqrt{2} dt \\ &= \frac{3\sqrt{2}}{2}. \quad \square \end{aligned}$$

4) In polar coordinates dl is given by $\sqrt{r^2 + \left(\frac{\partial r}{\partial \theta}\right)^2} d\theta$. To arrive this formula, one may either consider that θ is the parameter instead of t , and use the polar substitution, or that $(dl)^2 = (dr)^2 + (r d\theta)^2$.

In our case,

$$\mathcal{L} = \int_0^{2\pi} dl = \int_0^{2\pi} \sqrt{1 + \theta^2} d\theta.$$

5



Let C be the boundary curve for the solid S on the xy -plane.
That is, $C: x^2 + y^2 = 4$. And let

$$x(t) = 2\cos t, \quad y(t) = 2\sin t \quad \text{for } t \in [0, 2\pi]$$

be the parametrization of C . Then, $dl = 2dt$.

Now, let f be the height function given by $f(x, y, z) = z = x + y^2$. Then,

$f(\vec{r}(t)) = 2\cos t + 4\sin^2 t$. Then the surface area becomes

$$\int_0^{2\pi} (2\cos t + 4\sin^2 t) 2dt.$$

□

~~~~~  $\mathcal{M} \nabla$  ~~~~~

① Let  $\vec{F} = x\hat{i} + x^2y\hat{j}$ . Parametrize  $C = C_1 \cup C_2$  by

$C_1: x = 2\cos t, y = 2\sin t$  with  $t_1 = \pi, t_2 = \frac{\pi}{2}$  implies that  $dx = -2\sin t dt$   
 $dy = 2\cos t dt$ . Then under this parametrization, the integral becomes for part  $C_1$ ,

$$\int_{\pi}^{\pi/2} 2\cos t \cdot (-2\sin t) dt + 4\cos^2 t \cdot 2\sin t (2\cos t) dt.$$

Since  $C_2$  is a line segment on  $x=0$ , this part has zero contribution.

□

②  $C$  is a piecewise smooth, positively oriented, simple closed curve.

Let  $\vec{F} = x^2y\hat{i} + x^3y^2\hat{j}$ .  $\vec{F}$  has continuous first partial derivatives inside the region  $C$ .

Hence, we can apply Green's theorem.

$$\int_C \vec{F} \cdot d\vec{r} = \iint \nabla \times \vec{F} \, dA = \iint_{-2x^2}^{2x^2} (3x^2y^2 - x^2) dy dx.$$

□

$$(3) \vec{F}(\vec{r}(t)) = (\pi t^5 + e^t, 2\pi t^4 + 2t^2, t^3 + \sin(\pi t^3))$$

$$\dot{\vec{r}} = (1, 2t, 3\pi t^2) dt. \text{ Hence,}$$

$$\begin{aligned} \vec{F} \cdot \dot{\vec{r}} &= [(\pi t^5 + e^t) + (2\pi t^5 + 2t^3) + (3\pi t^5 + 3\pi t^2 \sin(\pi t^3))] dt \\ &= (8\pi t^5 + 2t^3 + e^t + 3\pi t^2 \sin(\pi t^3)) dt \end{aligned}$$

□

(5) Evaluate the line integral of the vector field  $\vec{F}$  over the curve  $C$ . Yes, by definition, because the potential function  $\phi$  is the simply connected domain over which  $\nabla \times \vec{F} = 0$  holds true.

(6) Evaluate  $\frac{1}{x^2+y^2}$  over the curve  $(x, y, z) = (0, 0, z)$  the function is defined over the simply connected domain. So, suppose there exists a potential  $\phi$ . Then it must satisfy

$$\frac{\partial \phi}{\partial x} = \frac{-2x}{x^2+y^2} \Rightarrow \phi = -\ln|x^2+y^2| + C_1(y, z).$$

$$\Rightarrow \frac{\partial \phi}{\partial y} = \frac{-2y}{x^2+y^2} + \frac{\partial C_1(y, z)}{\partial y} = z \cos(yz) - \frac{2y}{x^2+y^2} \Rightarrow \frac{\partial C_1(y, z)}{\partial y} = z \cos(yz)$$

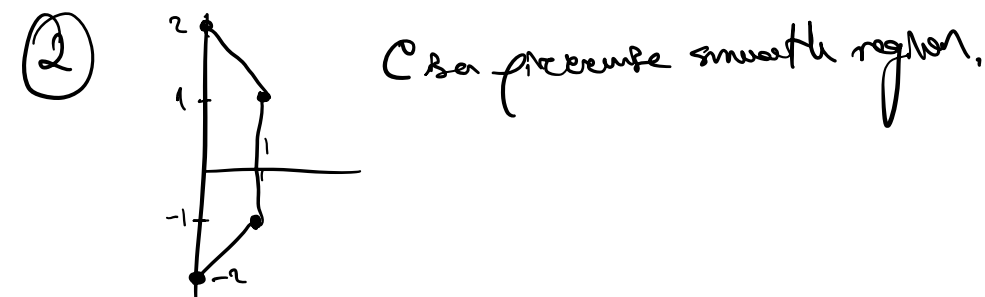
$$\Rightarrow C_1(y, z) = \sin(yz) + C_2(z). \Rightarrow \phi = -\ln|x^2+y^2| + \sin(yz) + C_2(z)$$

$$\Rightarrow \frac{\partial \phi}{\partial z} = y \cos(yz) + C_2'(z) = y \cos(yz) + 12z^2 \Rightarrow C_2(z) = 4z^3 + C.$$

$$\text{Hence, } \phi = -\ln|x^2+y^2| + \sin(yz) + 4z^3 + C.$$

□

~~~~~  $\mathcal{M}$  ~~~~~



EXAMPLE 1

Find the flux of the vector field $\mathbf{F} = m\mathbf{r}/|\mathbf{r}|^3$ out of a sphere $\mathcal{S}$ of radius a centred at the origin. (Here $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.)

Sol.:

$$\iint_{\mathcal{S}} \vec{F} \cdot d\vec{S} = \iint_{\mathcal{S}} m \frac{\vec{r}}{|\mathbf{r}|^3} \cdot \frac{\vec{r}}{|\mathbf{r}|} dS = \iint_{\mathcal{S}} m \frac{|\mathbf{r}|^2}{|\mathbf{r}|^4} dS \stackrel{*}{=} m \frac{1}{a^2} \iint_{\mathcal{S}} dS = 4\pi m.$$

(*) : since at the surface of the sphere $|\mathbf{r}| = a$. $\square$

EXAMPLE 2

Calculate the total flux of $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ outward through the surface of the solid cylinder $x^2 + y^2 \leq a^2$, $-h \leq z \leq h$.

Sol.:

$$\text{Top/Bottom: } \iint_{\mathcal{S}_{\text{top/bottom}}} \vec{F} \cdot d\vec{S} = \iint_{\mathcal{S}_{\text{top/bottom}}} (x\hat{i} + y\hat{j} + z\hat{k}) \cdot \hat{k} dS \stackrel{*}{=} \iint_{\mathcal{S}_{\text{top/bottom}}} h dS = \pi a^2 h.$$

(*) : since at the top of the cylinder $z = h$.

Sol.:

Using cylindrical coordinates, $\vec{F} = a \cos \theta \hat{i} + a \sin \theta \hat{j} + z \hat{k}$, $\hat{n} \stackrel{*}{=} \cos \theta \hat{i} + \sin \theta \hat{j}$,
 $dS = a d\theta dz$.

$$\text{Then, } \iint_{\mathcal{S}_{\text{side}}} \vec{F} \cdot d\vec{S} = \iint_{\mathcal{S}_{\text{side}}} a^2 d\theta dz = 4\pi a^2 h.$$

$$\text{Hence } \iint_{\mathcal{S}} \vec{F} \cdot d\vec{S} = 2\pi a^2 h + 4\pi a^2 h. \quad \square$$

EXAMPLE 3

Find the flux of $z\mathbf{i} + x^2\mathbf{k}$ upward through that part of the surface $z = x^2 + y^2$ lying above the square R defined by $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$.

Sol.:

Note that the surface is given by $\phi(x, y, z) = x^2 + y^2 - z = 0$ with one-to-one projection onto the domain R on the xy -plane. Hence, we may write

$$\vec{\partial S} = \frac{\nabla \phi}{\phi_z} dx dy = \frac{2x\hat{i} + 2y\hat{j} - \hat{k}}{-1} dx dy = (-2x\hat{i} - 2y\hat{j} + \hat{k}) dx dy.$$

Then,

$$\vec{F} \cdot \vec{\partial S} = (2xz - x^2) dx dy = (-2x^3 - 2xy^2 + x^2) dx dy,$$

and

$$\iint_S \vec{F} \cdot \vec{\partial S} = \iint_R (2x^3 + 2xy^2 - x^2) dx dy = \int_{-1}^1 dx \int_{-1}^1 (2x^3 - x^2 + 2xy^2) dy = \frac{4}{3}. \quad \square$$

EXAMPLE 5

Find the flux of $\mathbf{F} = \frac{2x\mathbf{i} + 2y\mathbf{j}}{x^2 + y^2} + \mathbf{k}$ downward through the surface S defined parametrically by

$$\mathbf{r} = u \cos v \mathbf{i} + u \sin v \mathbf{j} + u^2 \mathbf{k}, \quad (0 \leq u \leq 1, 0 \leq v \leq 2\pi).$$

Sol.:

$$\vec{\partial S} = \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv = (-2u^2 \cos v \hat{i} - 2u^2 \sin v \hat{j} + u \hat{k}) du dv.$$

Question asks for downward flux and coefficient for $\hat{k}$ is $u \geq 0$ on S . Hence, we shall use the opposite direction, thus,

$$\vec{\partial S} = (2u^2 \cos v \hat{i} + 2u^2 \sin v \hat{j} - u \hat{k}) du dv.$$

And on S , $\vec{F}$ becomes

$$\vec{F} = \frac{2x\hat{i} + 2y\hat{j}}{x^2 + y^2} + \hat{k} = \frac{2u \cos v \hat{i} + 2u \sin v \hat{j}}{u^2} + \hat{k}.$$

Hence, the outward flux of $\vec{F}$ through S is

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_V 3u \, du \, dv = 3\pi.$$

□

Vector differential identities

Let ϕ and ψ be scalar fields and $\mathbf{F}$ and $\mathbf{G}$ be vector fields, all assumed to be sufficiently smooth that all the partial derivatives in the identities are continuous. Then the following identities hold:

- (a) $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$
- (b) $\nabla \bullet (\phi\mathbf{F}) = (\nabla\phi) \bullet \mathbf{F} + \phi(\nabla \bullet \mathbf{F})$
- (c) $\nabla \times (\phi\mathbf{F}) = (\nabla\phi) \times \mathbf{F} + \phi(\nabla \times \mathbf{F})$
- (d) $\nabla \bullet (\mathbf{F} \times \mathbf{G}) = (\nabla \times \mathbf{F}) \bullet \mathbf{G} - \mathbf{F} \bullet (\nabla \times \mathbf{G})$
- (e) $\nabla \times (\mathbf{F} \times \mathbf{G}) = (\nabla \bullet \mathbf{G})\mathbf{F} + (\mathbf{G} \bullet \nabla)\mathbf{F} - (\nabla \bullet \mathbf{F})\mathbf{G} - (\mathbf{F} \bullet \nabla)\mathbf{G}$
- (f) $\nabla(\mathbf{F} \bullet \mathbf{G}) = \mathbf{F} \times (\nabla \times \mathbf{G}) + \mathbf{G} \times (\nabla \times \mathbf{F}) + (\mathbf{F} \bullet \nabla)\mathbf{G} + (\mathbf{G} \bullet \nabla)\mathbf{F}$
- (g) $\nabla \bullet (\nabla \times \mathbf{F}) = 0$ (div curl = 0)
- (h) $\nabla \times (\nabla\phi) = \mathbf{0}$ (curl grad = 0)
- (i) $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \bullet \mathbf{F}) - \nabla^2 \mathbf{F}$
(curl curl = grad div - Laplacian)

EXAMPLE 1

Show that the vector field $\mathbf{F} = (x^2 + yz)\mathbf{i} - 2y(x + z)\mathbf{j} + (xy + z^2)\mathbf{k}$ is solenoidal in $\mathbb{R}^3$ and find a vector potential for it.

Sol.:

We shall show that $\nabla \cdot \vec{F} = 0$ for every point in $\mathbb{R}^3$. So,

$$\begin{aligned} \nabla \cdot \vec{F} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}) \\ &= \frac{\partial}{\partial x} F_1 + \frac{\partial}{\partial y} F_2 + \frac{\partial}{\partial z} F_3 \end{aligned}$$

$$= (2x) + (-2(x+z)) + (2z) = 0 \quad \forall (x, y, z) \in \mathbb{R}^3. \text{ Hence } \vec{F} \text{ is solenoidal.}$$

Now, note that $\vec{F}$ is a smooth, solenoidal vector field on $\mathbb{R}^3$ and that $\mathbb{R}^3$ has a property that every closed surface in $\mathbb{R}^3$ bounds a domain contained in $\mathbb{R}^3$. Hence by theorem 5,

$\vec{F}$ has vector potential $\vec{\phi}$ of the form $\vec{F} = \nabla \times \vec{\phi}$. Once more note that $\vec{\phi}$ is not unique. We shall now find such a $\vec{\phi}$.

Since,

$$\nabla \times \vec{\phi} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ \phi_1 & \phi_2 & \phi_3 \end{vmatrix},$$

$\vec{\phi} = (\phi_1, \phi_2, \phi_3)$ must satisfy

$$(i) \quad \partial_y \phi_3 - \partial_z \phi_2 = x^2 + y^2,$$

$$(ii) \quad \partial_z \phi_1 - \partial_x \phi_3 = -2xy - 2yz$$

$$(iii) \quad \partial_x \phi_2 - \partial_y \phi_1 = z^2 + x^2$$

very related concept
with Gauge theory

Now note that ϕ has 3 independent first partial derivatives and there are 3 eq. with 6 unknowns presented above. This makes ϕ non-unique. Furthermore, we have left 6 degrees of freedom. So, we shall assume some terms to be equal some things for simplicity.

Eq. (i) and (iii) seems similar on the right hand-side and involves $\partial_x \phi_2$ on the left. So, we shall assume that $\phi_2 = 0$, by which there left 3 d.o.f.

Then eq. (i) implies that

$$\partial_y \phi_3 = x^2 + y^2 \Rightarrow \phi_3 = x^2 y + \frac{1}{2} y^2 z + C_1(x, z).$$

We may assume $C_1(x, z) = 0$, by which there left 1 d.o.f. So, we take

$$\phi_3 = x^2 y + \frac{1}{2} y^2 z.$$

From eq. (ii),

$$\partial_z \phi_1 = \partial_x \phi_3 - 2xy - 2yz = -2yz \Rightarrow \phi_1 = -y z^2 + C_2(x, y).$$

Since there remains 1 d.o.f., we cannot assume $C_2 = 0$.

From eq. (iii),

$$-\partial_y \phi_1 = x^2 + xy \stackrel{(ii)^*}{=} x^2 - \partial_y \phi_2 \Rightarrow \phi_2 = -\frac{1}{2}xy^2 + \phi_3(x).$$

We may now assume $\phi_3 = 0$.

Therefore, we find

$$\phi = -\left(yx^2 + \frac{1}{2}xy^2\right)\hat{i} + 0\hat{j} + \left(x^2y + \frac{1}{2}y^2x\right)\hat{k}.$$

□

Green's Theorem

Let R be a regular, closed region in the xy -plane whose boundary, $\mathcal{C}$, consists of one or more piecewise smooth, simple closed curves that are positively oriented with respect to R . If $\mathbf{F} = F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j}$ is a smooth vector field on R , then

$$\oint_{\mathcal{C}} F_1(x, y) dx + F_2(x, y) dy = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA.$$

$$\oint_{\mathcal{C}} x dy = -\oint_{\mathcal{C}} y dx = \frac{1}{2} \oint_{\mathcal{C}} x dy - y dx = \iint_R 1 dA = \text{area of } R.$$

EXAMPLE 2

Use the result of the previous example to calculate the area of the elliptic disk bounded by the curve $\mathcal{C}$ given by

$$\mathbf{r} = 3(\cos t + \sin t)\mathbf{i} + 2(\sin t - \cos t)\mathbf{j}, \quad 0 \leq t \leq 2\pi.$$

Sol.:

$$x = 3(\cos t + \sin t) \Rightarrow dx = 3(-\sin t + \cos t) dt$$

$$y = 2(\sin t - \cos t) \Rightarrow dy = 2(\cos t + \sin t) dt.$$

$$\text{Hence, } x dy - y dx = 12 dt. \text{ Thus, area} = \frac{1}{2} \oint_{\mathcal{C}} x dy - y dx = \frac{1}{2} \int_0^{2\pi} 12 dt = 12\pi.$$

The Divergence Theorem in the Plane

Let R be a regular, closed region in the xy -plane whose boundary, $\mathcal{C}$, consists of one or more piecewise smooth, simple closed curves. Let $\hat{\mathbf{N}}$ denote the unit outward (from R) normal field on $\mathcal{C}$. If $\mathbf{F} = F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j}$ is a smooth vector field on R , then

$$\iint_R \operatorname{div} \mathbf{F} dA = \oint_{\mathcal{C}} \mathbf{F} \cdot \hat{\mathbf{N}} ds.$$

~~... 14.2~~ we say the three-dimensional domain D is **x -simple** if it is bounded by a piecewise smooth surface $\mathcal{S}$ and if every straight line parallel to the x -axis and passing through an interior point of D meets $\mathcal{S}$ at exactly two points. Similar definitions hold for y -simple and z -simple, and we call the domain D **regular** if it is a union of finitely many, nonoverlapping subdomains, each of which is x -simple, y -simple, and z -simple.

The Divergence Theorem (Gauss's Theorem)

Let D be a regular, three-dimensional domain whose boundary $\mathcal{S}$ is an oriented, closed surface with unit normal field $\hat{\mathbf{N}}$ pointing out of D . If $\mathbf{F}$ is a smooth vector field defined on D , then

$$\iiint_D \operatorname{div} \mathbf{F} \, dV = \iint_{\mathcal{S}} \mathbf{F} \cdot \hat{\mathbf{N}} \, dS.$$

EXAMPLE 2

Evaluate $\iint_{\mathcal{S}} (x^2 + y^2) \, dS$, where $\mathcal{S}$ is the sphere $x^2 + y^2 + z^2 = a^2$. Use the Divergence Theorem.

Sol.:

$\vec{r} = \vec{\rho}$, hence $\hat{r} = \frac{\vec{\rho}}{a} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{a}$. we wish to find $\vec{F}$ such that $\vec{F} \cdot \hat{r} = x^2 + y^2$. Observe that $\vec{F} = a(x\hat{i} + y\hat{j})$ will satisfy. Hence, if $\mathcal{B}$ is the ball bounded by $\mathcal{S}$, then

$$\iint_{\mathcal{S}} (x^2 + y^2) \, dS = \iiint_{\mathcal{B}} \nabla \cdot \vec{F} \, dV = \iiint_{\mathcal{B}} 2a \, dV = 2a \left(\frac{4}{3} \pi a^3 \right).$$

EXAMPLE 3

By using the Divergence Theorem with $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, calculate the volume of a cone having base area A and height h . The base can be any smoothly bounded plane region.

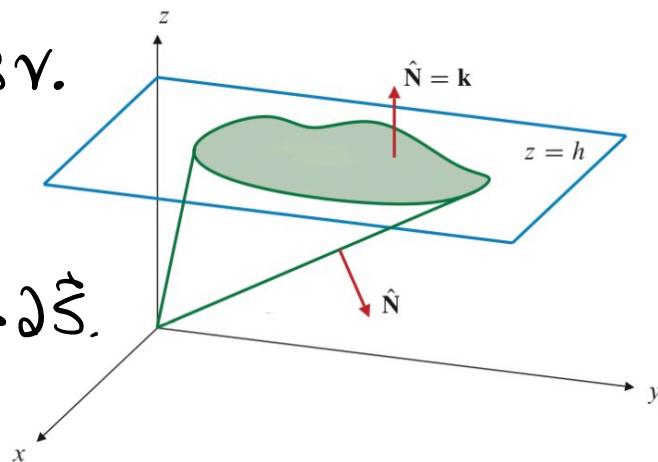
Sol.:

Let D be the solid cone ^{with $\mathcal{S} = \partial D$} consisting of the conical wall $\mathcal{W}$ and of the base $\mathcal{B}$. Without loss of generality, let the vertex of the cone be at origin and $\mathcal{B}$ is on the plane $z = h$.

First note that $\nabla \cdot \vec{F} = 3$, hence $\iiint_C \nabla \cdot \vec{F} \, dV = 3V$.

So,

$$3V = \iiint_C \nabla \cdot \vec{F} \, dV = \oiint_S \vec{F} \cdot d\vec{S} = \iint_W \vec{F} \cdot d\vec{S} + \iint_B \vec{F} \cdot d\vec{S}.$$



Note that $\vec{F} \cdot \hat{u} = 0$ over the surface W , since both $\vec{F}$ and position vector of W are parallel to each other and this unit normal vector $\hat{u}$ is perpendicular to $\vec{F}$ at any point on W . And over B , $\vec{F} \cdot \hat{u} = z = h$. Plus,

$$3V = 0 + h \iint_B dS = hA \Rightarrow V = \frac{1}{3}Ah.$$

□

EXAMPLE 4

Let $\mathcal{S}$ be the surface of an arbitrary regular domain D in 3-space that contains the origin in its interior. Find

$$\oiint_{\mathcal{S}} \mathbf{F} \cdot \hat{\mathbf{N}} \, dS,$$

where $\mathbf{F}(\mathbf{r}) = m\mathbf{r}/|\mathbf{r}|^3$ and $\hat{\mathbf{N}}$ is the unit outward normal on $\mathcal{S}$. (See Figure 16.13.)

Soln:

Since $\vec{F}$ is undefined at the origin which is an interior point of D , Divergence Theorem is not applicable.

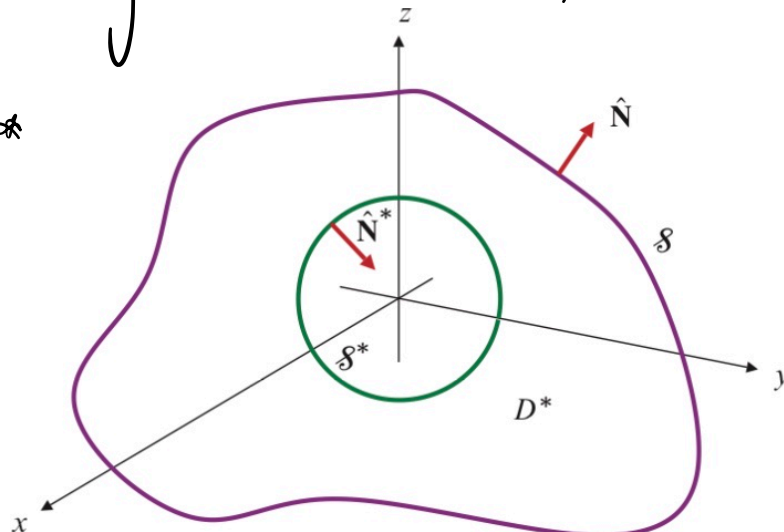
Let S^* be a small sphere centered at origin bounding a ball contained in D , and let D^* be the part of D that lies outside S^* .

We may now use the Div. Theorem over D^*

$$0 = \iiint_{D^*} \nabla \cdot \vec{F} \, dV = \oiint_{\mathcal{S}} \vec{F} \cdot \hat{u} \, dS + \underbrace{\oiint_{S^*} \vec{F} \cdot \hat{n}^* \, dS}_{=-4\pi m}$$

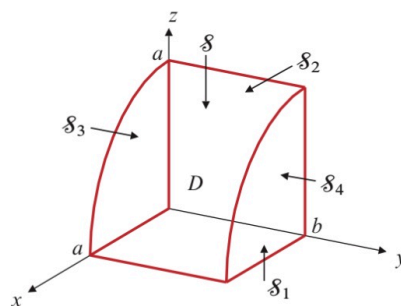
(*) : see previous computations.

Therefore, $\oiint_{\mathcal{S}} \vec{F} \cdot d\vec{S} = 4\pi m$.



EXAMPLE 5

Find the flux of $\mathbf{F} = x\mathbf{i} + y^2\mathbf{j} + z\mathbf{k}$ upward through the first octant part $\mathcal{S}$ of the cylindrical surface $x^2 + z^2 = a^2$, $0 \leq y \leq b$.



Sol.:

$$\begin{aligned} \oint_{\mathcal{S}} \vec{F} \cdot d\vec{S} &= \oint_S \vec{F} \cdot d\vec{S} + \oint_{S_4} \vec{F} \cdot d\vec{S} + \oint_{S_3} \vec{F} \cdot d\vec{S} + \oint_{S_2} \vec{F} \cdot d\vec{S} + \oint_{S_1} \vec{F} \cdot d\vec{S} = \iiint_D \nabla \cdot \vec{F} \, dV. \\ \underbrace{\oint_S}_{=I} \quad \underbrace{\oint_{S_4}}_{S_4: y=b} \quad \underbrace{\oint_{S_3}}_{S_3: y=0} \quad \underbrace{\oint_{S_2}}_{=0} \quad \underbrace{\oint_{S_1}}_{=0} \quad \underbrace{\iiint_D}_{=2V+2V\bar{y},} \\ &\quad \hat{n} = \hat{j} \Rightarrow \vec{F} \cdot \hat{n} = y^2 = b^2 \Rightarrow \oint_{S_4} \vec{F} \cdot \hat{n} = b^2 \oint_{S_4} dS = b^2 \oint_{S_4} dS = \frac{1}{4} \pi a^2 b^2 \\ &\quad \hat{n} = -\hat{j} \Rightarrow \vec{F} \cdot \hat{n} = -y^2 = 0 \Rightarrow \oint_{S_3} \vec{F} \cdot \hat{n} = 0. \\ &\quad \bar{y} = b/2 \Rightarrow = \frac{1}{2} \pi a^2 b \left(1 + \frac{b}{2}\right). \end{aligned}$$

$$\text{Thus, } I = \frac{1}{2} \pi a^2 b \left(1 + \frac{b}{2}\right) - \frac{1}{4} \pi a^2 b^2 = \frac{1}{2} \pi a^2 b. \quad \square$$

If D satisfies the conditions of the Divergence Theorem and has surface $\mathcal{S}$, and if $\mathbf{F}$ is a smooth vector field and ϕ is a smooth scalar field, then

$$(a) \quad \iiint_D \mathbf{curl} \, \mathbf{F} \, dV = - \iint_{\mathcal{S}} \mathbf{F} \times \hat{\mathbf{N}} \, dS,$$

$$(b) \quad \iiint_D \mathbf{grad} \, \phi \, dV = \iint_{\mathcal{S}} \phi \hat{\mathbf{N}} \, dS.$$

Stokes's Theorem

Let $\mathcal{S}$ be a piecewise smooth, oriented surface in 3-space, having unit normal field $\hat{\mathbf{N}}$ and boundary $\mathcal{C}$ consisting of one or more piecewise smooth, closed curves with orientation inherited from $\mathcal{S}$. If $\mathbf{F}$ is a smooth vector field defined on an open set containing $\mathcal{S}$, then

$$\oint_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r} = \iint_{\mathcal{S}} \mathbf{curl} \mathbf{F} \cdot \hat{\mathbf{N}} dS.$$

EXAMPLE 1 Evaluate $\oint_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = -y^3\mathbf{i} + x^3\mathbf{j} - z^3\mathbf{k}$, and $\mathcal{C}$ is the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $2x + 2y + z = 3$ oriented so as to have a counterclockwise projection onto the xy -plane.

Sol.:

Let $\mathcal{S}$ be the surface on the plane $2x + 2y + z - 3 = 0$ with boundary $\mathcal{C}$. Then,
 $d\vec{S} = (2\hat{i} + 2\hat{j} + \hat{k}) dx dy$. Also $\nabla \times \vec{F} = 3(x^2 + y^2)\hat{k}$. Hence,
 $\oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \iint_{\mathcal{S}} (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_{\mathcal{S}} 3(x^2 + y^2) dx dy = 3 \int_0^{2\pi} \int_0^1 r^2 r dr d\theta = \frac{3\pi}{2} \quad \square$

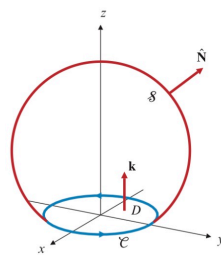
EXAMPLE 2 Find $I = \iint_{\mathcal{S}} \mathbf{curl} \mathbf{F} \cdot \hat{\mathbf{N}} dS$, where $\mathcal{S}$ is that part of the sphere $x^2 + y^2 + (z - 2)^2 = 8$ that lies above the xy -plane, $\hat{\mathbf{N}}$ is the unit outward normal field on $\mathcal{S}$, and

$$\mathbf{F} = y^2 \cos xz \mathbf{i} + x^3 e^{yz} \mathbf{j} - e^{-xyz} \mathbf{k}.$$

Sol.:

The very first thing that one should observe is that the curve $\mathcal{C}$ defined by $x^2 + y^2 = 4$ is a boundary for $\mathcal{S}$. Also, $\mathcal{C}$ is a boundary for the region $\mathcal{D}$ defined by $x^2 + y^2 \leq 4$ with $z = 0$. The rest of the solution is then straight forward.

$$I = \iint_{\mathcal{S}} (\nabla \times \vec{F}) \cdot d\vec{S} = \oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \iint_{\mathcal{D}} (\nabla \times \vec{F}) \cdot d\vec{S},$$



On $\mathcal{D}$:

$$\hat{n} = \hat{k},$$

$$(\nabla \times \vec{F}) \cdot \hat{k} = \left(\partial_x x^2 e^{y^2} - \partial_y y^2 \cos xz \right) \bigg|_{z=0}^{z=2\pi} = 3x^2 e^{y^2} - 2y \cos xz \bigg|_{z=0}^{z=2\pi} = 3x^2 - 2y.$$

Hence,

$$\iint_{\mathcal{D}} (3x^2 - 2y) dS = \iint_{\mathcal{D}} 3x^2 dA = 3 \int_0^{2\pi} \cos^2 \theta d\theta \int_0^2 r^3 dr = 12\pi.$$

(*) by symmetry, $\iint_{\mathcal{D}} y dA = 0.$

□

Let $\vec{r}$ be a position vector of a point P in xyz -space. In uvw -space,

$$\vec{r} = x(u, v, w)\hat{i} + y(u, v, w)\hat{j} + z(u, v, w)\hat{k}.$$

* $\frac{\partial \vec{r}}{\partial u} = \frac{\partial x}{\partial u}\hat{i} + \frac{\partial y}{\partial u}\hat{j} + \frac{\partial z}{\partial u}\hat{k}$ is tangent to the u -curve ($\vec{r} = \vec{r}(u, v_0, w_0)$)
and is normal to the u -surface.

* $\hat{u} = \frac{\partial \vec{r}}{\partial u} / \left\| \frac{\partial \vec{r}}{\partial u} \right\| = \frac{1}{h_u} \frac{\partial \vec{r}}{\partial u}.$

* $dV = h_u h_v h_w du dv dw$: volume element

* $dS_u = h_v h_w dv dw$: area element on u -surface

* $dl_u = h_u du$: arc-length element on u -curve

Let $f(u, v, w)$ and $\vec{F} = F_1 \hat{u} + F_2 \hat{v} + F_3 \hat{w}.$

* $\nabla f = \frac{1}{h_u} \frac{\partial f}{\partial u} \hat{u} + \dots + \dots.$

* $\nabla \cdot \vec{F} = \frac{1}{h_u h_v h_w} \left(\frac{\partial}{\partial u} (h_v h_w F_1) + \dots + \dots \right).$

* $\nabla \times \vec{F} = \frac{1}{h_u h_v h_w} \begin{vmatrix} h_u \hat{u} & h_v \hat{v} & h_w \hat{w} \\ \frac{\partial}{\partial u} & \frac{\partial}{\partial v} & \frac{\partial}{\partial w} \\ h_u F_1 & h_v F_2 & h_w F_3 \end{vmatrix}.$

* $\nabla^2 f = \nabla \cdot (\nabla f) = \frac{1}{h_u h_v h_w} \left(\frac{\partial}{\partial u} \left(h_v h_w \frac{1}{h_u} \frac{\partial f}{\partial u} \right) + \dots + \dots \right).$

Cylindrical

$$\vec{r} = r \cos \theta \hat{i} + r \sin \theta \hat{j} + z \hat{k}$$

$$h_r = 1, h_\theta = r, h_z = 1$$

Spherical

$$\vec{r} = R \sin \phi \cos \theta \hat{i} + R \sin \phi \sin \theta \hat{j} + R \cos \phi \hat{k}$$

$$h_R = 1, h_\phi = R, h_\theta = R \sin \phi$$